Sequences — nth terms · The difference method · Arithmetic sequences and series

Calculator allowed. Marks in brackets. ★ = harder question. $S_n = \frac{n}{2}\left[2a + (n-1)d\right]$.

1 Write down the first three terms of the sequence whose nth term is [3]

a $5n + 2$ **b** $n^2 - n$ **c** $\dfrac{2n+1}{n+2}$

2 Use the difference method to find the next three terms of each sequence. [3]

a $3,\ 7,\ 15,\ 27,\ 43,\ \ldots$ **b** $20,\ 17,\ 14,\ 11,\ \ldots$ **c** $1,\ 2,\ 4,\ 7,\ 11,\ \ldots$

3 Find a formula for the nth term of each sequence. [4]

a $6,\ 10,\ 14,\ 18,\ \ldots$ **b** $93,\ 86,\ 79,\ 72,\ \ldots$

4 The 3rd term of an arithmetic sequence is 11 and the 8th term is 31. Find the first term a and the common difference d. [4]

5 The 6th term of an arithmetic sequence is 23 and the 15th term is 50. Find the sum of the first 30 terms. [5]

6 The sum of the first 24 terms of an arithmetic sequence is 1080. The common difference is 2. Find the first term. [3]

7 Logs are stacked in rows. The bottom row has 30 logs, each row above has one log fewer than the row below, and the top row has 6 logs. How many logs are in the stack? [3]

Total: 25 marks

Check: a constant first difference a means the nth term is $an + b$ — find b from the first term · arithmetic sequence: nth term $a + (n-1)d$ · two given terms → two equations, subtract to find d · $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$ · count the number of terms carefully (last − first + 1 when $d = \pm 1$).